How Does Family Involvement Influence Student Academic Performance?

Classification Project

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Loading Data and Necessary Packages.

```
library(tidyverse)
library(tidymodels)
library(readxl)
library(discrim)
library(nnet)
library(glmnet)
library(vip)
library(broom)
library(kableExtra)
library(patchwork)
library(forcats)
library(themis)
library(poissonreg)
```

We begin by importing the curated 2019 PFI dataset and inspecting its structure.

```
pfi_data <- read_excel("~/STA 631 01/projects/pfi-data.xlsx", sheet = "curated 2019")
dim(pfi_data)
```

```
## [1] 15500    75
```

```
names(pfi_data)
```

```
##  [1] "BASMID"       "ALLGRADEX"    "EDCPUB"       "SCCHOICE"     "SPUBCHOIX"
##  [6] "SCONSIDR"     "SCHLHRSWK"    "EINTNET"      "MOSTIMPT"     "INTNUM"
## [11] "SEENJOY"      "SEGRADES"     "SEABSNT"      "SEGRADEQ"     "FSSPORTX"
## [16] "FSVOL"        "FSMTNG"       "FSPTMTNG"     "FSATCNFN"     "FSFUNDRS"
## [21] "FSCOMMTE"     "FSCOUNSLR"    "FSFREQ"       "FSNOTESX"     "FSMEMO"
## [26] "FCSCHOOL"     "FCTEACHR"     "FCSTDS"       "FCORDER"      "FCSUPPRT"
## [31] "FHHOME"       "FHWKHRS"      "FHAMOUNT"     "FHCAMT"       "FHPLACE"
## [36] "FHCHECKX"     "FHHELP"       "FOSTORY2X"    "FOCRAFTS"     "FOGAMES"
## [41] "FOBUILDX"     "FOSPORT"      "FORESPON"     "FOHISTX"      "FODINNERX"
## [46] "FOLIBRAYX"    "FOBOOKSTX"    "HDHEALTH"     "CDOBMM"       "CDOBYY"
## [51] "CSEX"         "CSPEAKX"      "HHTOTALXX"    "RELATION"     "P1REL"
## [56] "P1SEX"        "P1MRSTA"      "P1EMPL"       "P1HRSWK"      "P1MTHSWRK"
## [61] "P1AGE"        "P2GUARD"      "TTLHHINC"     "OWNRNTHB"     "CHLDNT"
## [66] "SEFUTUREX"    "DSBLTY"       "HHPARN19X"    "HHPARN19_BRD" "NUMSIBSX"
## [71] "PARGRADEX"    "RACEETH"      "INTACC"       "CENREG"       "ZIPLOCL"
```

```r
glimpse(pfi_data)
```

```
## Rows: 15,500
## Columns: 75
## $ BASMID       <dbl> 2.0191e+10, 2.0191e+10, 2.0191e+10, 2.0191e+10, 2.0191e+1~
## $ ALLGRADEX    <dbl> 12, 4, 11, 6, 9, 6, 8, 5, 9, 2, 15, 12, 6, 14, 6, 9, 2, 7~
## $ EDCPUB       <dbl> 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, ~
## $ SCCHOICE     <dbl> 2, 2, 2, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 2, 1, ~
## $ SPUBCHOIX    <dbl> 3, 3, 2, 1, 3, 2, 1, 1, 1, 1, 3, 3, 2, 2, 2, 1, 2, 2, 1, ~
## $ SCONSIDR     <dbl> 2, 2, 1, 1, 2, 2, 2, 1, 2, 2, 1, 1, 2, 2, 2, 2, 1, 2, 1, ~
## $ SCHLHRSWK    <dbl> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, ~
## $ EINTNET      <dbl> 4, 3, 3, 4, 4, 4, 4, 4, 4, 4, 3, 4, 4, 4, 4, 4, 4, 4, 4, ~
## $ MOSTIMPT     <dbl> -1, -1, 2, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1~
## $ INTNUM       <dbl> -1, 2, 2, -1, -1, -1, -1, -1, -1, -1, 6, -1, -1, -1, -1, ~
## $ SEENJOY      <dbl> 2, 3, 3, 1, 2, 1, 1, 1, 2, 2, 1, 3, 3, 2, 1, 2, 1, 2, 2, ~
## $ SEGRADES     <dbl> 1, 5, 3, 1, 1, 5, 1, 5, 1, 5, 2, 1, 2, 2, 2, 1, 5, 1, 3, ~
## $ SEABSNT      <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 3, ~
## $ SEGRADEQ     <dbl> 1, 3, 4, 1, 1, 1, 2, 3, 1, 3, 3, 2, 4, 2, 2, 2, 1, 1, 3, ~
## $ FSSPORTX     <dbl> 2, 2, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, ~
## $ FSVOL        <dbl> 2, 1, 1, 1, 2, 2, 2, 2, 1, 1, 2, 1, 2, 2, 2, 2, 1, 2, 1, ~
## $ FSMTNG       <dbl> 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 1, 2, 1, 1, 1, 1, 1, ~
## $ FSPTMTNG     <dbl> 2, 2, 1, 1, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 1, ~
## $ FSATCNFN     <dbl> 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 1, 2, 1, 1, 1, 1, 1, ~
## $ FSFUNDRS     <dbl> 1, 2, 2, 1, 1, 1, 2, 1, 1, 1, 2, 1, 2, 2, 1, 2, 1, 1, 1, ~
## $ FSCOMMTE     <dbl> 2, 2, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, ~
## $ FSCOUNSLR    <dbl> 2, 2, 1, 1, 2, 2, 2, 2, 2, 2, 2, 1, 1, 1, 2, 1, 2, 2, 1, ~
## $ FSFREQ       <dbl> 2, 2, 20, 3, 5, 5, 3, 5, 2, 4, 1, 3, 7, 2, 5, 4, 3, 4, 10~
## $ FSNOTESX     <dbl> 2, 1, 2, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, ~
## $ FSMEMO       <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ FCSCHOOL     <dbl> 2, 1, 4, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 2, 1, 2, 2, ~
## $ FCTEACHR     <dbl> 2, 2, 3, 1, 1, 1, 2, 1, 1, 1, 2, 1, 1, 1, 1, 2, 1, 3, 2, ~
## $ FCSTDS       <dbl> 1, 1, 4, 1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 1, 1, 2, 1, 1, 1, ~
## $ FCORDER      <dbl> 2, 1, 4, 1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 1, 1, 3, 2, 2, 3, ~
## $ FCSUPPRT     <dbl> 1, 1, 4, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 1, 3, ~
## $ FHHOME       <dbl> 4, 4, 2, 3, 3, 3, 3, 3, 4, 4, 2, 3, 4, 4, 3, 2, 6, 1, 3, ~
## $ FHWKHRS      <dbl> 2, 7, 6, 5, 5, 3, 10, 2, 10, 4, 5, 3, 3, 24, 5, 8, -1, 1,~
## $ FHAMOUNT     <dbl> 1, 2, 3, 1, 3, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, -1, 3, 1,~
## $ FHCAMT       <dbl> 1, 2, 2, 1, 1, 1, 1, 1, 1, 2, 2, 1, 2, 1, 2, 2, -1, 1, 1,~
## $ FHPLACE      <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 1, -1, 2, 1,~
## $ FHCHECKX     <dbl> 3, 4, 3, 4, 4, 4, 4, 4, 3, 4, 4, 2, 3, 4, 3, 3, -1, 4, 3,~
## $ FHHELP       <dbl> 2, 4, 1, 3, 3, 3, 2, 3, 1, 4, 2, 2, 3, 2, 2, 2, -1, 1, 2,~
## $ FOSTORY2X    <dbl> 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 1, 1, 1, 1, ~
## $ FOCRAFTS     <dbl> 1, 2, 2, 1, 2, 2, 2, 1, 2, 1, 2, 2, 2, 2, 1, 2, 1, 1, 2, ~
## $ FOGAMES      <dbl> 2, 2, 2, 1, 1, 2, 1, 1, 1, 1, 1, 2, 2, 2, 1, 2, 1, 1, 1, ~
## $ FOBUILDX     <dbl> 1, 2, 1, 1, 1, 2, 1, 2, 2, 1, 2, 2, 2, 2, 2, 1, 1, 2, 1, ~
## $ FOSPORT      <dbl> 2, 1, 1, 1, 1, 1, 2, 1, 2, 1, 1, 2, 2, 1, 1, 1, 1, 1, 1, ~
## $ FORESPON     <dbl> 1, 2, 1, 1, 1, 1, 1, 2, 1, 2, 1, 1, 2, 1, 2, 2, 2, 1, 1, ~
## $ FOHISTX      <dbl> 1, 1, 1, 1, 1, 1, 2, 1, 2, 2, 2, 1, 2, 1, 1, 2, 1, 1, 1, ~
## $ FODINNERX    <dbl> 7, 5, 7, 7, 3, 7, 7, 7, 7, 4, 0, 2, 1, 7, 5, 7, 3, 7, 3, ~
## $ FOLIBRAYX    <dbl> 2, 2, 2, 1, 2, 2, 2, 1, 1, 1, 2, 2, 1, 2, 2, 2, 1, 1, 1, ~
## $ FOBOOKSTX    <dbl> 2, 2, 2, 1, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 1, 1, 2, 1, ~
## $ HDHEALTH     <dbl> 1, 1, 2, 1, 1, 1, 1, 1, 1, 2, 2, 1, 2, 2, 1, 2, 1, 1, 1, ~
## $ CDOBMM       <dbl> 1, 8, 3, 12, 5, 5, 12, 3, 4, 1, 12, 3, 3, 3, 1, 11, 1, 1,~
## $ CDOBYY       <dbl> 2004, 2012, 2005, 2009, 2006, 2010, 2007, 2011, 2007, 201~
```

```
## $ CSEX         <dbl> 2, 1, 1, 1, 1, 2, 2, 2, 1, 1, 2, 1, 1, 1, 2, 1, 2, 2, 2, ~
## $ CSPEAKX      <dbl> 2, 4, 2, 4, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 2, 2, 2, 2, 2, ~
## $ HHTOTALXX    <dbl> 4, 9, 3, 3, 4, 3, 5, 6, 4, 3, 5, 4, 3, 5, 4, 5, 6, 3, 3, ~
## $ RELATION     <dbl> 1, 1, 1, 1, 2, 2, 1, 5, 2, 1, 1, 1, 2, 1, 5, 1, 1, 1, 2, ~
## $ P1REL        <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ P1SEX        <dbl> 2, 2, 2, 2, 1, 1, 2, 2, 1, 2, 2, 2, 1, 2, 1, 2, 2, 2, 1, ~
## $ P1MRSTA      <dbl> 1, 5, 5, 4, 1, 1, 1, 1, 1, 1, 5, 1, 1, 1, 3, 1, 1, 1, 3, ~
## $ P1EMPL       <dbl> 3, 1, 1, 1, 2, 1, 3, 4, 1, 1, 1, 1, 1, 5, 1, 5, 2, 5, 1, ~
## $ P1HRSWK      <dbl> -1, 40, 35, 35, 60, 60, -1, -1, 40, 37, 36, 40, 60, -1, 4~
## $ P1MTHSWRK    <dbl> 6, 12, 12, 11, 12, 9, 6, 0, 12, 12, 12, 12, 12, 0, 12, 0,~
## $ P1AGE        <dbl> 52, 29, 33, 35, 48, 46, 34, 32, 49, 44, 38, 45, 32, 39, 3~
## $ P2GUARD      <dbl> 1, 2, 2, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 2, ~
## $ TTLHHINC     <dbl> 8, 5, 3, 2, 10, 10, 3, 12, 11, 7, 3, 10, 8, 5, 8, 8, 12, ~
## $ OWNRNTHB     <dbl> 2, 2, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 2, ~
## $ CHLDNT       <dbl> 1, 1, 4, 2, 2, 2, 1, 2, 1, 3, 1, 1, 3, 2, 2, 3, 2, 2, 1, ~
## $ SEFUTUREX    <dbl> 6, 6, 5, 6, 6, 6, 6, 5, 5, 3, 2, 5, 3, 5, 5, 3, 6, 5, 5, ~
## $ DSBLTY       <dbl> 2, 2, 2, 2, 2, 2, 1, 1, 2, 2, 1, 1, 1, 2, 2, 1, 2, 2, 2, ~
## $ HHPARN19X    <dbl> 1, 2, 2, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 3, 1, 1, 1, 3, ~
## $ HHPARN19_BRD <dbl> 1, 2, 2, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 2, ~
## $ NUMSIBSX     <dbl> 1, 1, 1, 1, 1, 0, 2, 1, 1, 0, 2, 1, 0, 2, 0, 2, 2, 0, 1, ~
## $ PARGRADEX    <dbl> 3, 3, 5, 2, 5, 5, 3, 4, 5, 3, 4, 5, 4, 2, 3, 3, 5, 4, 4, ~
## $ RACEETH      <dbl> 4, 3, 2, 3, 5, 1, 3, 1, 1, 1, 3, 1, 1, 3, 1, 1, 5, 1, 3, ~
## $ INTACC       <dbl> 1, 1, 3, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 3, 1, 1, 1, 1, 3, ~
## $ CENREG       <dbl> 4, 4, 2, 2, 2, 4, 2, 3, 3, 3, 3, 1, 3, 2, 1, 3, 1, 2, 4, ~
## $ ZIPLOCL      <dbl> 11, 12, 11, 31, 21, 21, 41, 41, 21, 42, 13, 21, 21, 21, 4~
```

Research Question

This project examines how family involvement at school and at home is associated with student academic performance in the 2019 Parent and Family Involvement (PFI) survey. Academic performance is measured using `SEGRADES`, the parent-reported overall grades item. Family involvement is measured using school-based involvement items, contact with the school, and homework/home-engagement variables.

Select Variables + Response (SEGRADES)

```r
pfi_clean <- pfi_data %>%
  mutate(across(where(is.numeric), ~ ifelse(. < 0, NA, .))) %>%
  select(
    BASMID,
    SEGRADES, SEENJOY, SEABSNT, SEGRADEQ,
    FSSPORTX, FSVOL, FSMTNG, FSPTMTNG, FSATCNFN, FSFUNDRS, FSCOMMTE, FSCOUNSLR, FSFREQ,
    FSNOTESX, FSMEMO,
    FHHOME, FHWKHRS, FHAMOUNT, FHCAMT, FODINNERX,
    EDCPUB, SCCHOICE, SCONSIDR, SPUBCHOIX,
    ALLGRADEX, PARGRADEX, TTLHHINC, RACEETH,
    CSEX, DSBLTY, HHPARN19_BRD, OWNRNTHB, CENREG,
    HHTOTALXX, NUMSIBSX, P1AGE, P1EMPL, INTACC, ZIPLOCL
  ) %>%
  mutate(
    SEGRADES = ifelse(SEGRADES == 5, NA, SEGRADES),
    grade_multi = factor(
      SEGRADES,
      levels = c(1, 2, 3, 4),
      labels = c("Mostly_A", "Mostly_B", "Mostly_C", "Mostly_D_or_lower")
```

Table 1: Top 20 Variables by Missingness

variable	n_missing	percent_missing
SEGRADES	1967	12.7
grade_multi	1967	12.7
high_achiever	1967	12.7
FHWKHRS	1047	6.8
FHAMOUNT	1047	6.8
FHCAMT	1047	6.8
FSSPORTX	126	0.8
FSVOL	126	0.8
FSMTNG	126	0.8
FSPTMTNG	126	0.8
FSATCNFN	126	0.8
FSFUNDRS	126	0.8
FSCOMMTE	126	0.8
FSCOUNSLR	126	0.8
FSFREQ	126	0.8
FSNOTESX	53	0.3
FSMEMO	53	0.3
FHHOME	53	0.3
BASMID	0	0.0
SEENJOY	2	0.0

```
    ),
    high_achiever = factor(
      ifelse(SEGRADES == 1, "Yes", "No"),
      levels = c("Yes", "No")
    )
  )
```

Lets check for missingness

```
missingness_table <- pfi_clean %>%
  summarise(across(everything(), ~ sum(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "n_missing") %>%
  mutate(percent_missing = round(100 * n_missing / nrow(pfi_clean), 1)) %>%
  arrange(desc(percent_missing))
```

```
missingness_table %>%
  slice_head(n = 20) %>%
  kbl(caption = "Top 20 Variables by Missingness") %>%
  kable_classic(full_width = FALSE)
```

```
pfi_model <- pfi_clean %>%
  filter(
    !is.na(grade_multi),
    !is.na(high_achiever),
    !is.na(PARGRADEX),
    !is.na(TTLHHINC),
    !is.na(RACEETH),
    !is.na(CSEX)
```

```r
  )
dim(pfi_model)
```

```
## [1] 13533    42
```

Lets add composite family-involvement variables

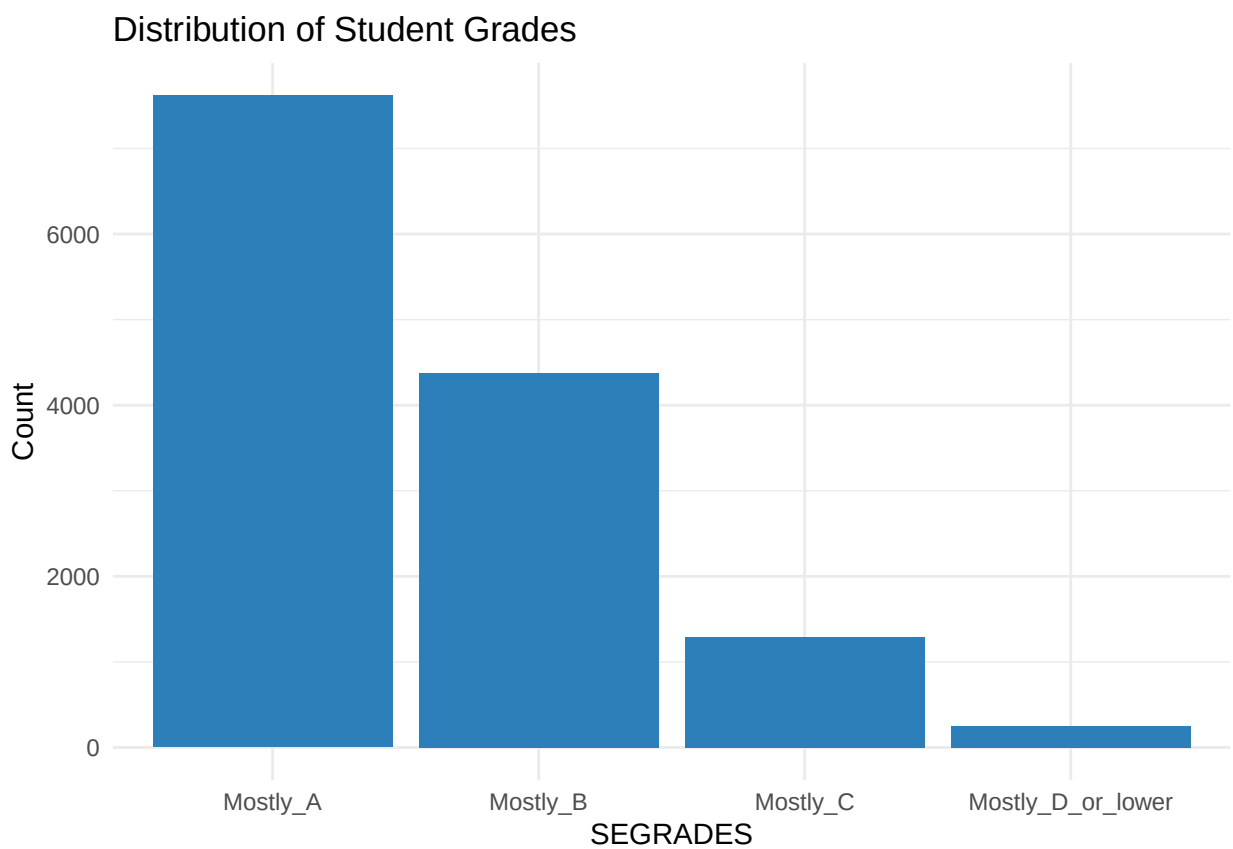
The questionnaire confirms those `FS*` variables are all school-based family involvement items, and `FSFREQ` is the count of how many times an adult in the household went to meetings or activities at the child's school.

```r
pfi_model <- pfi_model %>%
  mutate(
    across(
      c(FSSPORTX, FSVOL, FSMTNG, FSPTMTNG, FSATCNFN, FSFUNDRS, FSCOMMTE, FSCOUNSLR,
        FSNOTESX, FSMEMO),
      ~ case_when(
        . == 1 ~ 1,
        . == 2 ~ 0,
        TRUE ~ NA_real_
      )
    ),
    school_involve_score = rowSums(
      select(., FSSPORTX, FSVOL, FSMTNG, FSPTMTNG, FSATCNFN, FSFUNDRS, FSCOMMTE, FSCOUNSLR),
      na.rm = TRUE
    ),
    school_contact_score = rowSums(
      select(., FSNOTESX, FSMEMO),
      na.rm = TRUE
    )
  )
```

```r
pfi_model <- pfi_model %>%
  mutate(
    CSEX = factor(CSEX, levels = c(1, 2), labels = c("Male", "Female")),
    RACEETH = factor(RACEETH, levels = c(1, 2, 3, 4, 5),
                     labels = c("White", "Black", "Hispanic", "Asian_PI", "Other")),
    PARGRADEX = factor(PARGRADEX,
                       labels = c("Less_than_HS", "HS", "Associate", "Bachelors", "Graduate")),
    TTLHHINC = factor(TTLHHINC, ordered = TRUE),
    EDCPUB = factor(EDCPUB),
    SCCHOICE = factor(SCCHOICE),
    SCONSIDR = factor(SCONSIDR),
    SPUBCHOIX = factor(SPUBCHOIX),
    DSBLTY = factor(DSBLTY),
    HHPARN19_BRD = factor(HHPARN19_BRD),
    OWNRNTHB = factor(OWNRNTHB),
    CENREG = factor(CENREG),
    P1EMPL = factor(P1EMPL),
    INTACC = factor(INTACC),
    ZIPLOCL = factor(ZIPLOCL),
    SEENJOY = factor(SEENJOY, ordered = TRUE),
    SEABSNT = factor(SEABSNT, ordered = TRUE)
  )
```

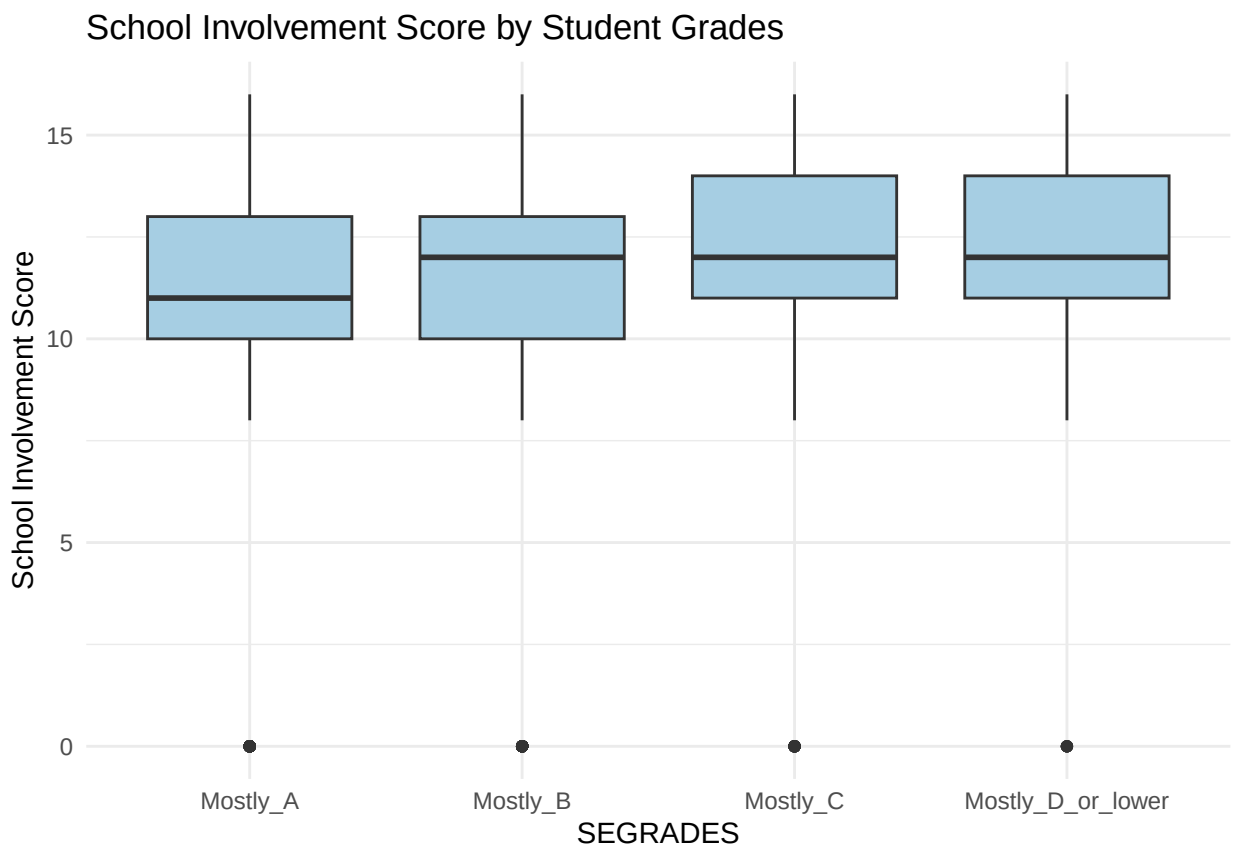
Exploratory Data Analysis

```
ggplot(pfi_model, aes(x = grade_multi)) +
  geom_bar(fill = "#2C7FB8") +
  labs(
    title = "Distribution of Student Grades",
    x = "SEGRADES",
    y = "Count"
  ) +
  theme_minimal()
```



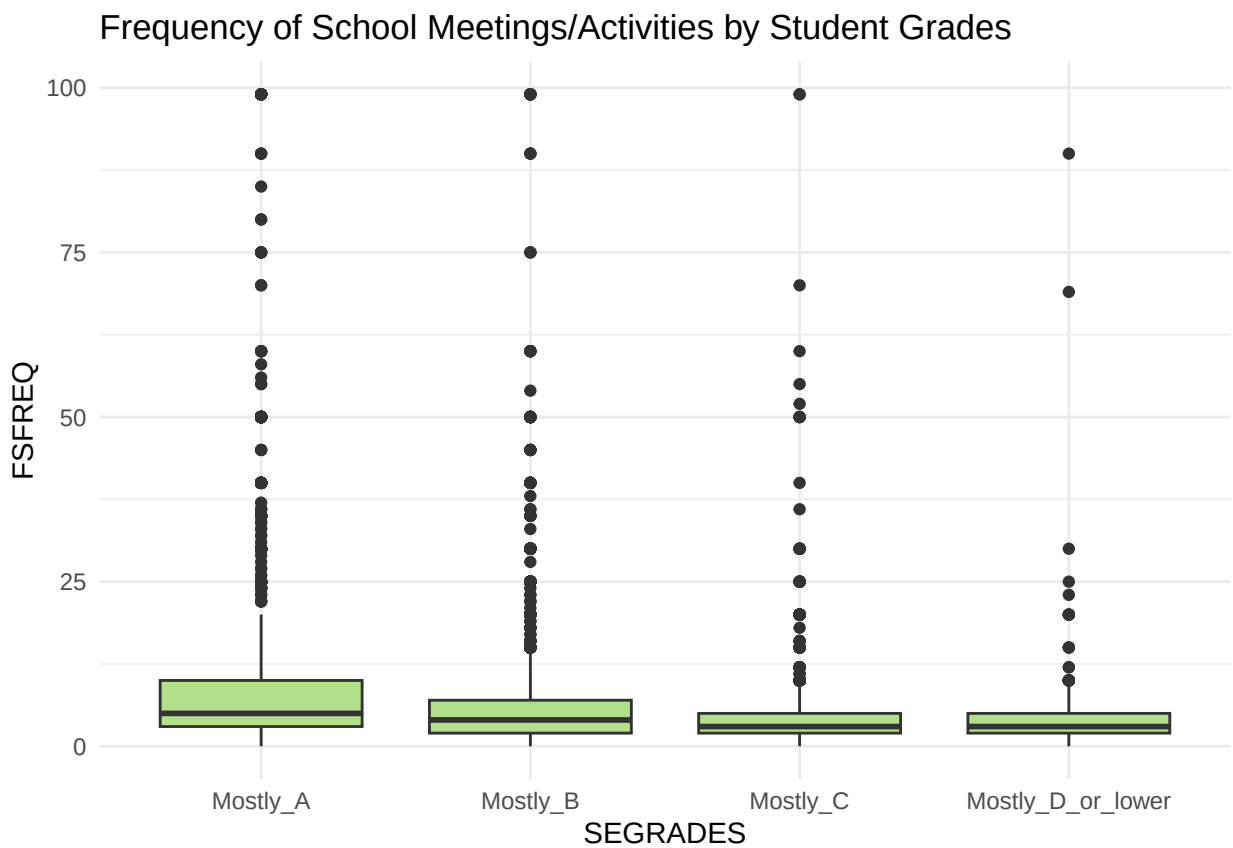
The distribution of SEGRADES shows a strong imbalance, with most students reporting mostly A's or B's and relatively few reporting lower grades. This imbalance is important for model selection and evaluation.

```
ggplot(pfi_model, aes(x = grade_multi, y = school_involve_score)) +
  geom_boxplot(fill = "#A6CEE3") +
  labs(
    title = "School Involvement Score by Student Grades",
    x = "SEGRADES",
    y = "School Involvement Score"
  ) +
  theme_minimal()
```



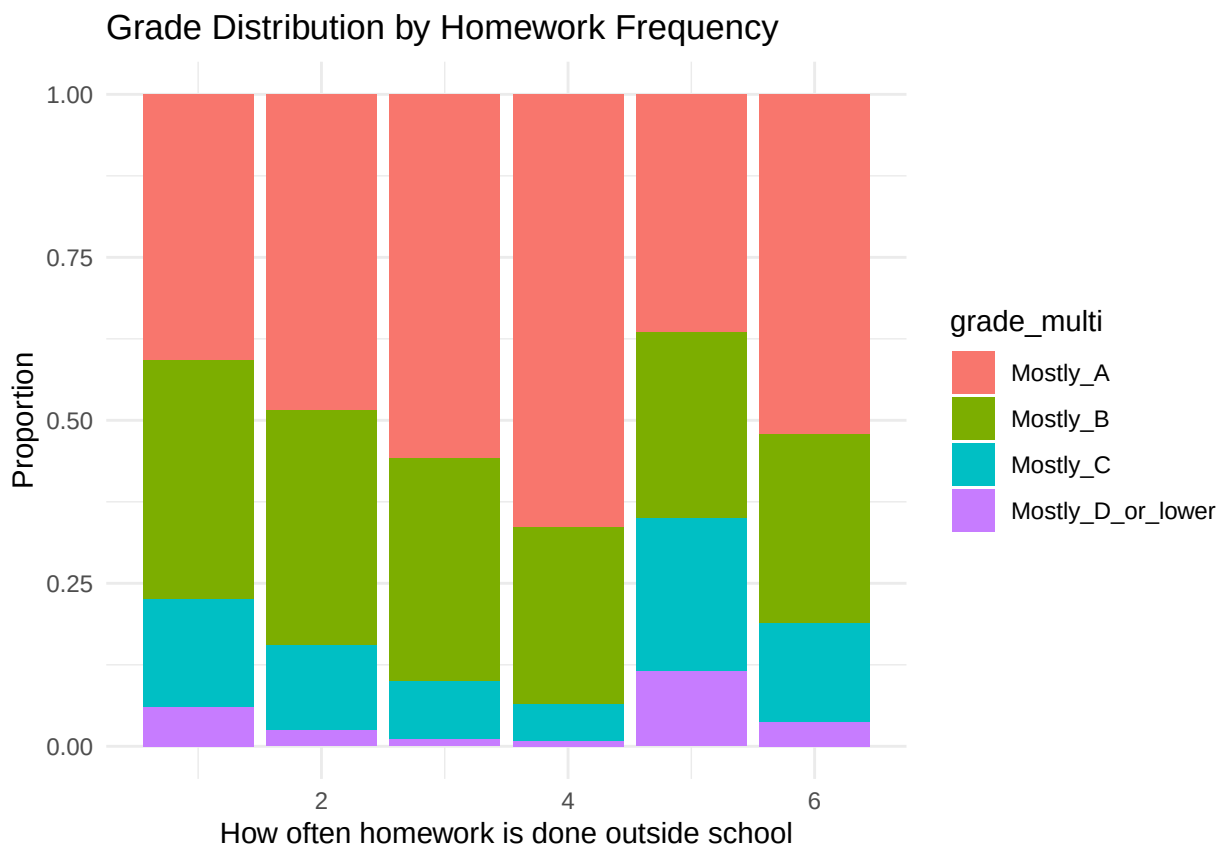
School involvement scores do not show a strong positive relationship with higher academic performance; in fact, students with lower grades often have similar or slightly higher involvement levels, suggesting that involvement may be reactive rather than preventative.

```
ggplot(pfi_model, aes(x = grade_multi, y = FSFREQ)) +
  geom_boxplot(fill = "#B2DF8A") +
  labs(
    title = "Frequency of School Meetings/Activities by Student Grades",
    x = "SEGRADES",
    y = "FSFREQ"
  ) +
  theme_minimal()
```



The frequency of school meetings and activities shows a weak positive association with academic performance, but the large variability and presence of extreme values suggest that this relationship is not strong and that the distribution is highly skewed.

```
ggplot(pfi_model, aes(x = FHHOME, fill = grade_multi)) +
  geom_bar(position = "fill") +
  labs(
    title = "Grade Distribution by Homework Frequency",
    x = "How often homework is done outside school",
    y = "Proportion"
  ) +
  theme_minimal()
```



Homework frequency shows a clear positive relationship with academic performance, with higher frequencies associated with a larger proportion of students receiving mostly A's. This suggests that home-based engagement may be a stronger predictor of academic success than school-based involvement.

Data splitting for modeling

```
set.seed(631)

class_df <- pfi_model %>%
  select(
    high_achiever, grade_multi, SEGRADES,
    school_involve_score, school_contact_score, FSFREQ,
    FHHOME, FHWKHRS, FHAMOUNT, FHCAMT, FODINNERX,
    SEENJOY, SEABSNT,
    EDCPUB, SCCHOICE, SCONSIDR, SPUBCHOIX,
    PARGRADEX, TTLHHINC, RACEETH, CSEX, DSBLTY,
    HHPARN19_BRD, OWNRNTHB, CENREG, HHTOTALXX, NUMSIBSX, P1AGE, P1EMPL, INTACC, ZIPLOCL
  ) %>%
  drop_na()

class_split <- initial_split(class_df, prop = 0.75, strata = high_achiever)
class_train <- training(class_split)
class_test  <- testing(class_split)
```

```r
set.seed(631)
class_folds <- vfold_cv(class_train, v = 10, strata = high_achiever)

count_df <- pfi_model %>%
  select(
    FSFREQ,
    school_involve_score, school_contact_score,
    FHHOME, FHWKHRS, FODINNERX,
    PARGRADEX, TTLHHINC, RACEETH, CSEX, DSBLTY,
    SCCHOICE, EDCPUB
  ) %>%
  drop_na()

count_split <- initial_split(count_df, prop = 0.75)
count_train <- training(count_split)
count_test  <- testing(count_split)
```

```r
multi_df <- class_df %>%
  select(
    grade_multi,
    school_involve_score, school_contact_score, FSFREQ,
    FHHOME, FHWKHRS, FODINNERX, SEENJOY, SEABSNT,
    SCCHOICE, EDCPUB, PARGRADEX, TTLHHINC, RACEETH, CSEX, DSBLTY
  ) %>%
  drop_na()

set.seed(631)
multi_split <- initial_split(multi_df, prop = 0.75, strata = grade_multi)

multi_train <- training(multi_split)
multi_test  <- testing(multi_split)
```

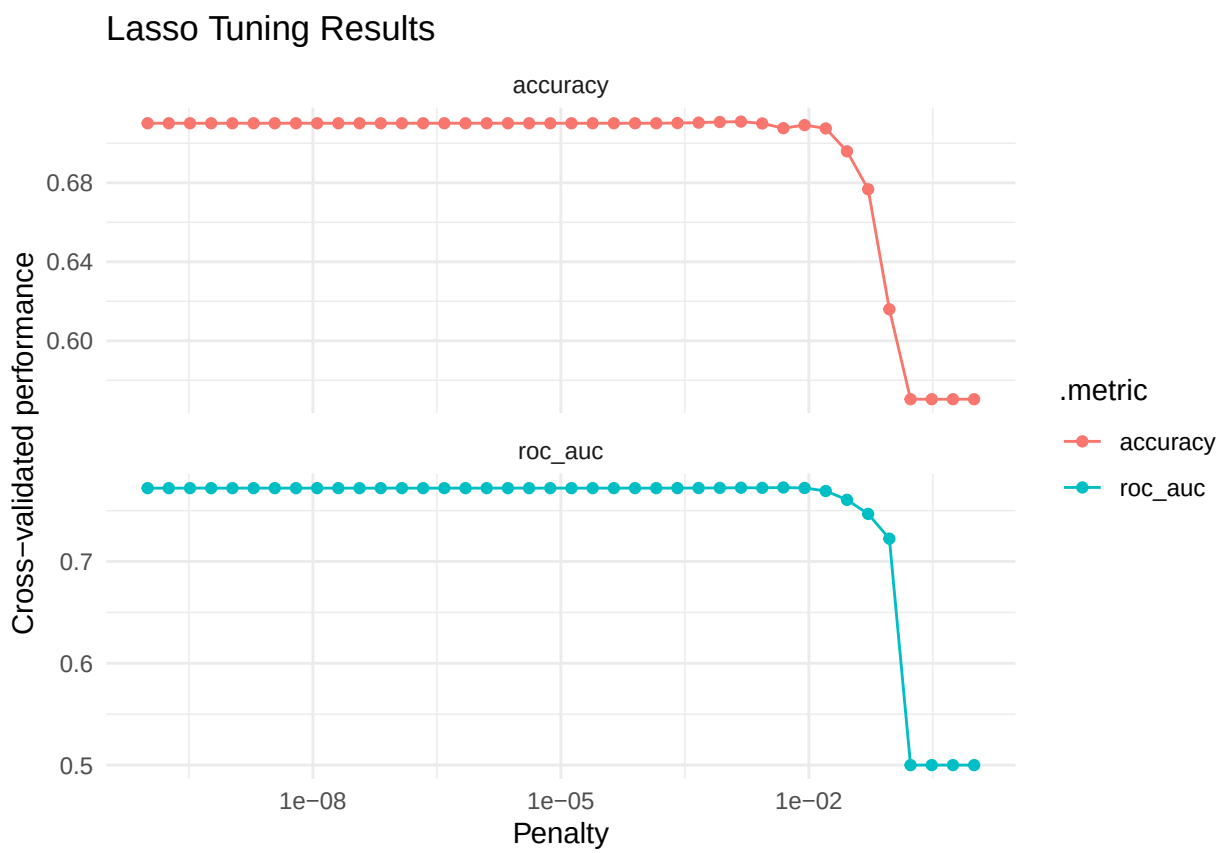
Lasso Regression

```r
lasso_recipe <- recipe(high_achiever ~ ., data = class_train) %>%
  step_rm(grade_multi, SEGRADES) %>%
  step_dummy(all_nominal_predictors()) %>%
  step_zv(all_predictors()) %>%
  step_normalize(all_numeric_predictors())
```

```r
lasso_spec <- logistic_reg(penalty = tune(), mixture = 1) %>%
  set_engine("glmnet") %>%
  set_mode("classification")

lasso_wf <- workflow() %>%
  add_recipe(lasso_recipe) %>%
  add_model(lasso_spec)

lambda_grid <- grid_regular(penalty(), levels = 40)

set.seed(631)
lasso_res <- tune_grid(
```

```r
  lasso_wf,
  resamples = class_folds,
  grid = lambda_grid,
  metrics = metric_set(roc_auc, accuracy),
  control = control_grid(save_pred = TRUE)
)
```

```r
collect_metrics(lasso_res) %>%
  ggplot(aes(x = penalty, y = mean, color = .metric)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ .metric, scales = "free_y", ncol = 1) +
  scale_x_log10() +
  theme_minimal() +
  labs(title = "Lasso Tuning Results", x = "Penalty", y = "Cross-validated performance")
```




```r
best_penalty <- select_best(lasso_res, metric = "roc_auc")

final_lasso_wf <- finalize_workflow(lasso_wf, best_penalty)

lasso_last_fit <- last_fit(final_lasso_wf, class_split)

lasso_preds <- collect_predictions(lasso_last_fit)
collect_metrics(lasso_last_fit)
```

```
## # A tibble: 3 x 4
##   .metric     .estimator .estimate .config
```

```
##   <chr>       <chr>          <dbl> <chr>
## 1 accuracy    binary         0.715 pre0_mod0_post0
## 2 roc_auc     binary         0.773 pre0_mod0_post0
## 3 brier_class binary         0.191 pre0_mod0_post0
```

The lasso logistic model achieved an accuracy of 71.5% and an ROC AUC of 0.773 on the test set, indicating moderate predictive ability in distinguishing high-achieving students from others. Performance remained stable across small penalty values but declined sharply at larger penalties, suggesting that overly aggressive regularization removed too much predictive information.

```r
final_lasso_fit <- fit(final_lasso_wf, data = class_train)

lasso_coefs <- final_lasso_fit %>%
  extract_fit_parsnip() %>%
  tidy()

lasso_coefs_nonzero <- lasso_coefs %>%
  filter(term != "(Intercept)", estimate != 0) %>%
  arrange(desc(abs(estimate)))

lasso_coefs_nonzero %>%
  slice_head(n = 20) %>%
  knitr::kable(caption = "Top Lasso-Selected Predictors")
```

Table 2: Top Lasso-Selected Predictors

term	estimate	penalty
SEENJOY_1	0.4124879	0.0049239
DSBLTY_X2	-0.3599257	0.0049239
PARGRADEX_Graduate	-0.3021157	0.0049239
CSEX_Female	-0.2390815	0.0049239
TTLHHINC_01	-0.2087388	0.0049239
RACEETH_Black	0.1939437	0.0049239
RACEETH_Hispanic	0.1761597	0.0049239
SEENJOY_2	-0.1480184	0.0049239
FHHOME	-0.1448516	0.0049239
PARGRADEX_Bachelors	-0.1344068	0.0049239
FHCAMT	0.1291649	0.0049239
SEABSNT_1	0.0760875	0.0049239
HHPARN19_BRD_X2	0.0718702	0.0049239
P1AGE	0.0649976	0.0049239
RACEETH_Asian_PI	-0.0640073	0.0049239
CENREG_X4	0.0613216	0.0049239
PARGRADEX_HS	0.0566932	0.0049239
CENREG_X2	-0.0492400	0.0049239
school_contact_score	-0.0451594	0.0049239
ZIPLOCL_X32	-0.0356906	0.0049239

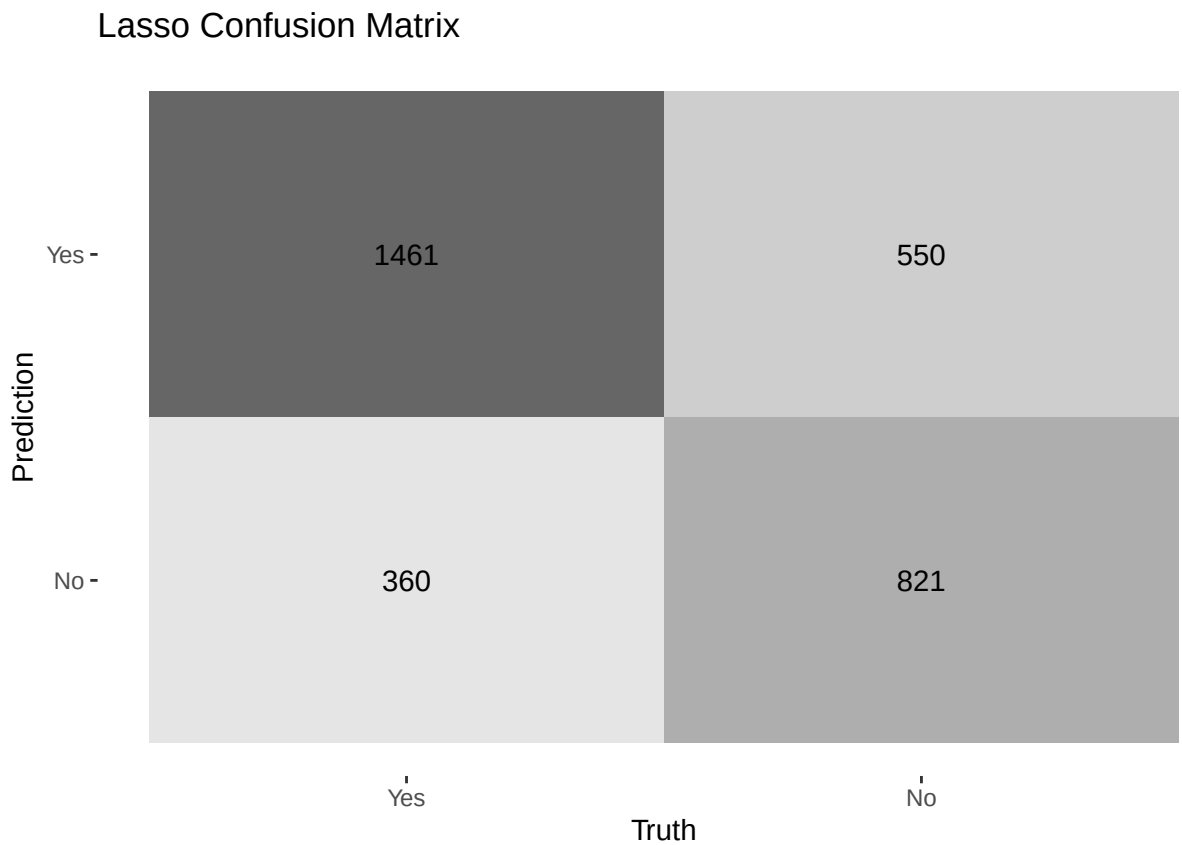
The lasso model selected a relatively small set of predictors, indicating that only a subset of the available family, school, and demographic variables contributed meaningful predictive information. Among the retained predictors, student school experience variables such as school enjoyment and absenteeism, along with household background characteristics such as parental education, income, disability status, and demographic factors, showed the strongest effects. Importantly for the research question, home-based involvement variables such as homework-related engagement (FHHOME, FHCAMT) remained in the model, while school contact was

retained with a smaller coefficient. This suggests that family involvement at home may be more strongly associated with academic performance than school-based involvement, although broader student and household characteristics also play a major role.

race/ethnicity remained in the selected model, indicating differences in predicted academic performance across groups after accounting for other retained covariates.

Confusion Matrix for lasso:

```
lasso_preds %>%
  conf_mat(truth = high_achiever, estimate = .pred_class) %>%
  autoplot(type = "heatmap") +
  labs(title = "Lasso Confusion Matrix")
```



Standard Logistic Regression Model

```
log_formula <- high_achiever ~ school_involve_score + school_contact_score + FSFREQ +
  FHHOME + FHWKHRS + FODINNERX + SEENJOY + SEABSNT +
  SCCHOICE + EDCPUB + PARGRADEX + TTLHHINC + RACEETH + CSEX + DSBLTY

log_recipe <- recipe(log_formula, data = class_train) %>%
  step_dummy(all_nominal_predictors()) %>%
  step_zv(all_predictors())

log_spec <- logistic_reg() %>%
```

```r
  set_engine("glm") %>%
  set_mode("classification")

log_wf <- workflow() %>%
  add_recipe(log_recipe) %>%
  add_model(log_spec)

log_last_fit <- last_fit(log_wf, class_split)
log_preds <- collect_predictions(log_last_fit)

collect_metrics(log_last_fit)
```

```
## # A tibble: 3 x 4
##   .metric     .estimator .estimate .config
##   <chr>       <chr>          <dbl> <chr>
## 1 accuracy    binary         0.708 pre0_mod0_post0
## 2 roc_auc     binary         0.769 pre0_mod0_post0
## 3 brier_class binary         0.193 pre0_mod0_post0
```

The standard logistic regression model achieved an accuracy of 70.8%, an ROC AUC of 0.769, and a Brier score of 0.193 on the test set. These results were very similar to the lasso logistic model, though the lasso model performed slightly better on all three metrics. This indicates that reducing the predictor set through regularization did not harm predictive performance and may have improved out-of-sample generalization. Because the lasso model is also more parsimonious, it provides a strong candidate for the preferred binary classification model

```r
metric_set(accuracy, precision, recall, f_meas, roc_auc)(
  log_preds,
  truth = high_achiever,
  estimate = .pred_class,
  .pred_Yes
)
```

```
## # A tibble: 5 x 3
##   .metric   .estimator .estimate
##   <chr>     <chr>          <dbl>
## 1 accuracy  binary         0.708
## 2 precision binary         0.723
## 3 recall    binary         0.790
## 4 f_meas    binary         0.755
## 5 roc_auc   binary         0.769
```

The higher recall relative to precision suggests that the model is better at identifying true positive cases than at avoiding false positive classifications.

```r
log_fit_object <- fit(log_wf, data = class_train)

tidy(extract_fit_parsnip(log_fit_object), conf.int = TRUE, exponentiate = TRUE) %>%
  kbl(caption = "Logistic Regression Odds Ratios") %>%
  kable_classic(full_width = FALSE)
```

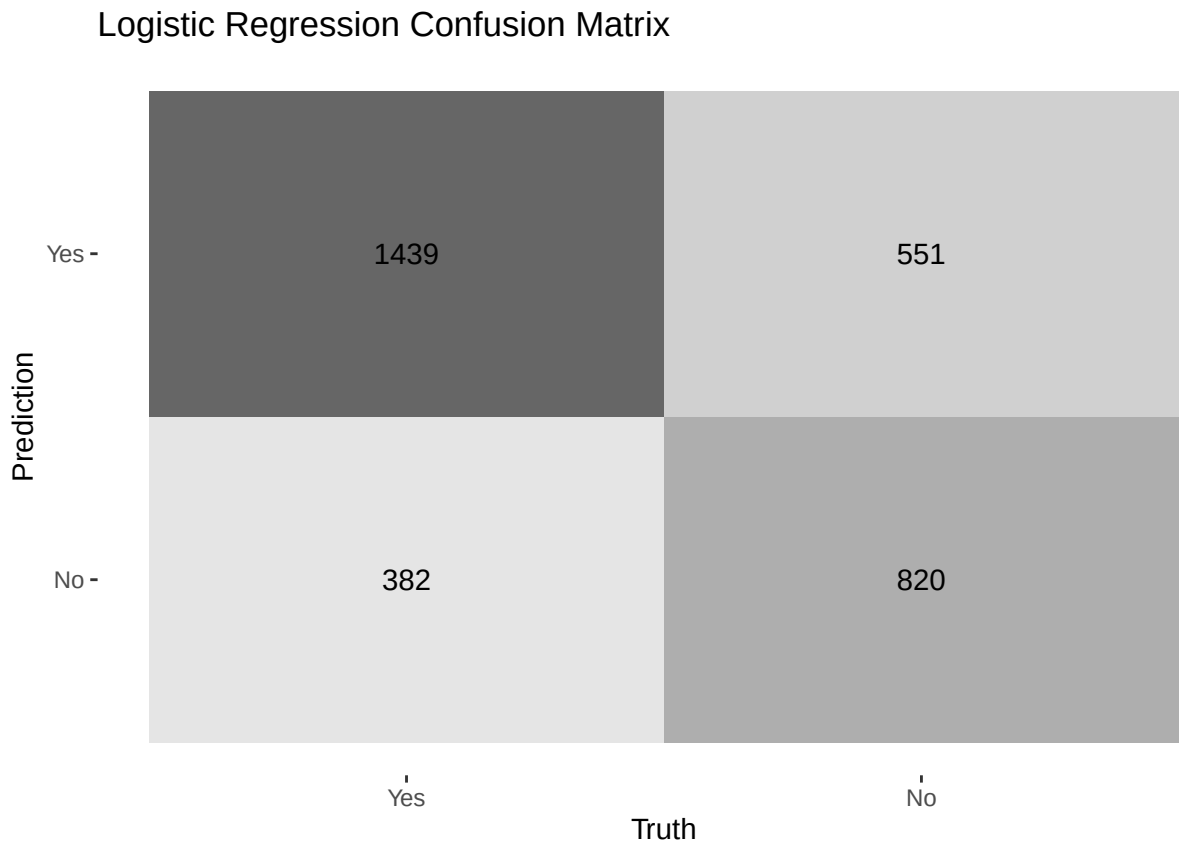
Confusion Matrix for logistic:

```r
log_preds %>%
  conf_mat(truth = high_achiever, estimate = .pred_class) %>%
```

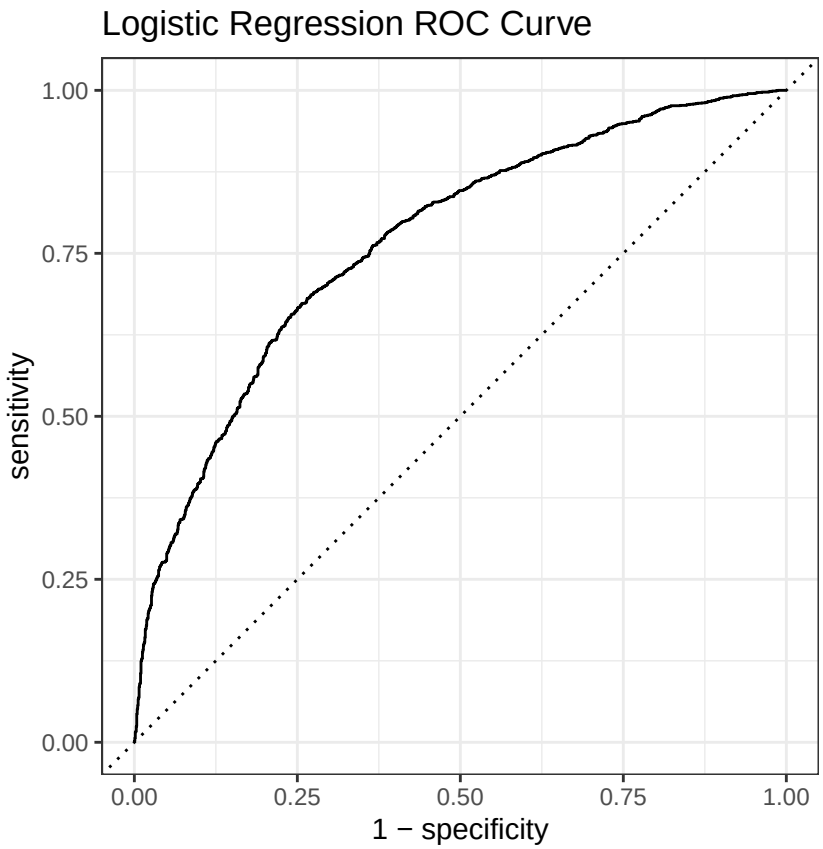
Table 3: Logistic Regression Odds Ratios

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	13.1729145	0.2618645	9.8454082	0.0000000	7.8978791	22.0481314
school_involve_score	1.0156403	0.0136713	1.1351735	0.2563026	0.9887850	1.0432257
school_contact_score	0.8664530	0.0398616	-3.5961306	0.0003230	0.8012676	0.9367968
FSFREQ	0.9979975	0.0027483	-0.7293740	0.4657729	0.9925799	1.0033392
FHHOME	0.8546945	0.0306294	-5.1261533	0.0000003	0.8048561	0.9075457
FHWKHRS	0.9995538	0.0050493	-0.0883976	0.9295606	0.9896971	1.0094923
FODINNERX	0.9694966	0.0117818	-2.6293311	0.0085553	0.9473583	0.9921416
SEENJOY_1	3.8365627	0.1278157	10.5196527	0.0000000	2.9982910	4.9524917
SEENJOY_2	0.6062330	0.1025449	-4.8807028	0.0000011	0.4970689	0.7434014
SEENJOY_3	0.8658592	0.0700573	-2.0559320	0.0397891	0.7547893	0.9934586
SEABSNT_1	1.5679110	0.1913098	2.3508677	0.0187297	1.0863142	2.3050736
SEABSNT_2	1.0533276	0.1571315	0.3306424	0.7409146	0.7780781	1.4426190
SEABSNT_3	1.1609103	0.1153429	1.2935728	0.1958130	0.9262671	1.4562678
SCCHOICE_X2	1.0077543	0.0498453	0.1549681	0.8768465	0.9138807	1.1110985
EDCPUB_X2	0.9523333	0.0814732	-0.5994636	0.5488637	0.8112800	1.1166176
PARGRADEX_HS	0.9298361	0.1261089	-0.5768581	0.5640353	0.7255995	1.1897746
PARGRADEX_Associate	0.7165266	0.1183000	-2.8177500	0.0048361	0.5676212	0.9026967
PARGRADEX_Bachelors	0.5016405	0.1231958	-5.5997968	0.0000000	0.3936177	0.6380956
PARGRADEX_Graduate	0.3470300	0.1281318	-8.2598099	0.0000000	0.2696862	0.4457235
TTLHHINC_01	0.4411260	0.1195522	-6.8457514	0.0000000	0.3488658	0.5574590
TTLHHINC_02	1.0302578	0.0989166	0.3013557	0.7631433	0.8487545	1.2508639
TTLHHINC_03	1.0206128	0.0902301	0.2261243	0.8211047	0.8548084	1.2175973
TTLHHINC_04	0.8815719	0.0944977	-1.3338814	0.1822427	0.7325351	1.0610139
TTLHHINC_05	1.1176032	0.0949471	1.1710357	0.2415844	0.9282033	1.3468123
TTLHHINC_06	0.8368187	0.0935647	-1.9040075	0.0569092	0.6966818	1.0054220
TTLHHINC_07	1.1341144	0.0896348	1.4040528	0.1603031	0.9516037	1.3522888
TTLHHINC_08	0.9868376	0.0842962	-0.1571808	0.8751023	0.8365292	1.1641288
TTLHHINC_09	0.9354899	0.0786952	-0.8473820	0.3967822	0.8017291	1.0914664
TTLHHINC_10	1.0105759	0.0755992	0.1391597	0.8893240	0.8713862	1.1719888
TTLHHINC_11	0.9343384	0.0812248	-0.8361565	0.4030669	0.7968118	1.0955985
RACEETH_Black	2.1653251	0.0845965	9.1324155	0.0000000	1.8351176	2.5568759
RACEETH_Hispanic	1.6536256	0.0625065	8.0466858	0.0000000	1.4630556	1.8693184
RACEETH_Asian_PI	0.7581050	0.1056804	-2.6204810	0.0087806	0.6149963	0.9308290
RACEETH_Other	1.0780326	0.0972970	0.7722503	0.4399662	0.8902733	1.3038168
CSEX_Female	0.5877427	0.0475344	-11.1806698	0.0000000	0.5354051	0.6450757
DSBLTY_X2	0.3943715	0.0569492	-16.3384592	0.0000000	0.3526254	0.4408335

```
  autoplot(type = "heatmap") +
  labs(title = "Logistic Regression Confusion Matrix")
```




```
log_preds %>%
  roc_curve(truth = high_achiever, .pred_Yes) %>%
  autoplot() +
  labs(title = "Logistic Regression ROC Curve")
```



Multinomial Model

```
multi_recipe <- recipe(grade_multi ~ ., data = multi_train) %>%
  step_dummy(all_nominal_predictors()) %>%
  step_zv(all_predictors())

multi_spec <- multinom_reg() %>%
  set_engine("nnet") %>%
  set_mode("classification")

multi_wf <- workflow() %>%
  add_recipe(multi_recipe) %>%
  add_model(multi_spec)

multi_last_fit <- last_fit(multi_wf, multi_split)

collect_metrics(multi_last_fit)
```

```
## # A tibble: 3 x 4
##   .metric     .estimator .estimate .config
##   <chr>       <chr>          <dbl> <chr>
## 1 accuracy    multiclass     0.624 pre0_mod0_post0
## 2 roc_auc     hand_till      0.712 pre0_mod0_post0
## 3 brier_class multiclass     0.239 pre0_mod0_post0
```

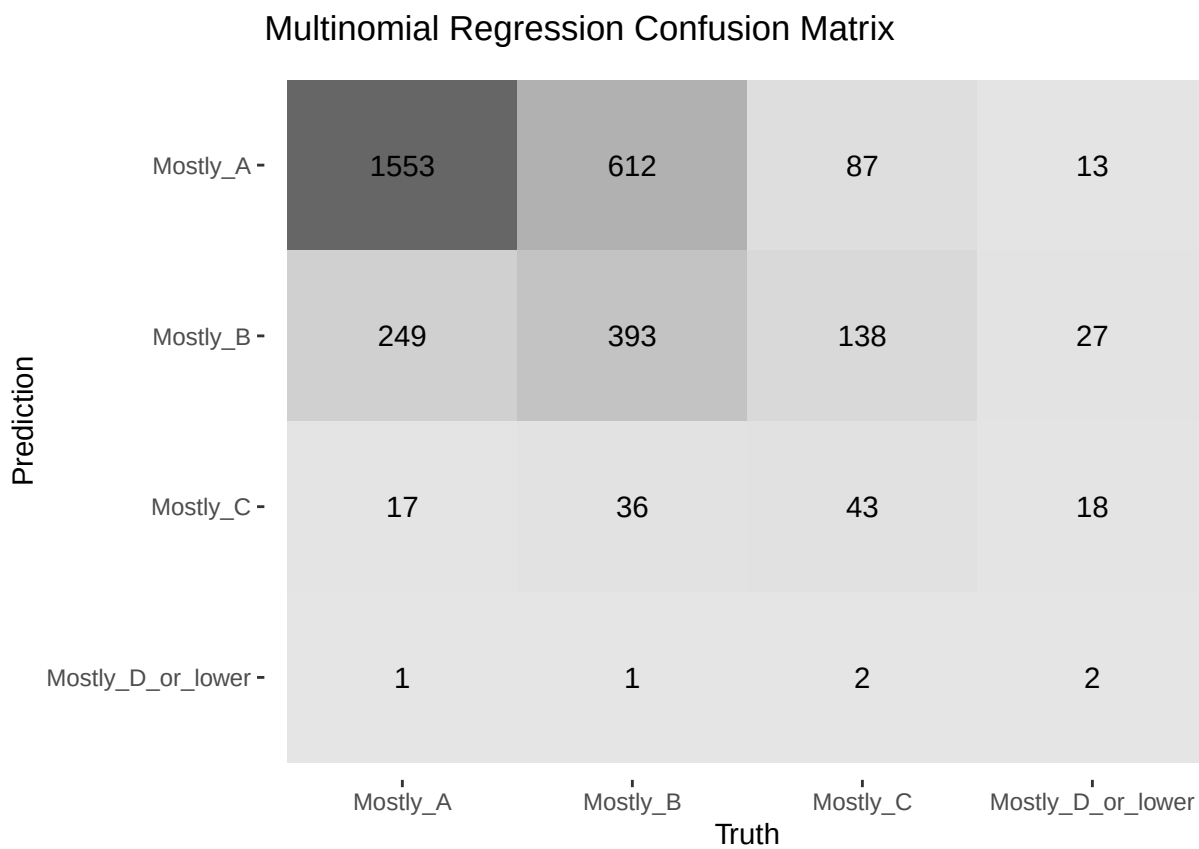
The multinomial model is harder to fit well than the binary models, but it is the more appropriate model for the full SEGRADES response because it preserves all four academic-performance categories instead of collapsing them into a yes/no outcome.

Multinomial Confusion Matrix

```
multi_preds <- collect_predictions(multi_last_fit)

multi_cm <- multi_preds %>%
  conf_mat(truth = grade_multi, estimate = .pred_class)

multi_cm
```

```
##                     Truth
## Prediction           Mostly_A Mostly_B Mostly_C Mostly_D_or_lower
##   Mostly_A               1553      612       87                 13
##   Mostly_B                249      393      138                 27
##   Mostly_C                 17       36       43                 18
##   Mostly_D_or_lower         1        1        2                  2
```

```
autoplot(multi_cm, type = "heatmap") +
  labs(title = "Multinomial Regression Confusion Matrix")
```



The multinomial confusion matrix shows that the model performs best on the more common A and B categories and struggles more with the less common C and D-or-lower categories, which is consistent with the class imbalance observed in the outcome distribution.

Linear and Quadratic Discriminant Analysis

```r
disc_recipe <- recipe(
  high_achiever ~ school_involve_score + school_contact_score + FSFREQ +
    FHWKHRS + FODINNERX + P1AGE + HHTOTALXX + NUMSIBSX,
  data = class_train
) %>%
  step_normalize(all_numeric_predictors()) %>%
  step_zv(all_predictors())

lda_spec <- discrim_linear() %>%
  set_engine("MASS") %>%
  set_mode("classification")

qda_spec <- discrim_quad() %>%
  set_engine("MASS") %>%
  set_mode("classification")

lda_wf <- workflow() %>% add_recipe(disc_recipe) %>% add_model(lda_spec)
qda_wf <- workflow() %>% add_recipe(disc_recipe) %>% add_model(qda_spec)

lda_fit <- last_fit(lda_wf, class_split)
qda_fit <- last_fit(qda_wf, class_split)

collect_metrics(lda_fit)
```

```
## # A tibble: 3 x 4
##   .metric     .estimator .estimate .config
##   <chr>       <chr>          <dbl> <chr>
## 1 accuracy    binary         0.607 pre0_mod0_post0
## 2 roc_auc     binary         0.617 pre0_mod0_post0
## 3 brier_class binary         0.236 pre0_mod0_post0
```

```r
collect_metrics(qda_fit)
```

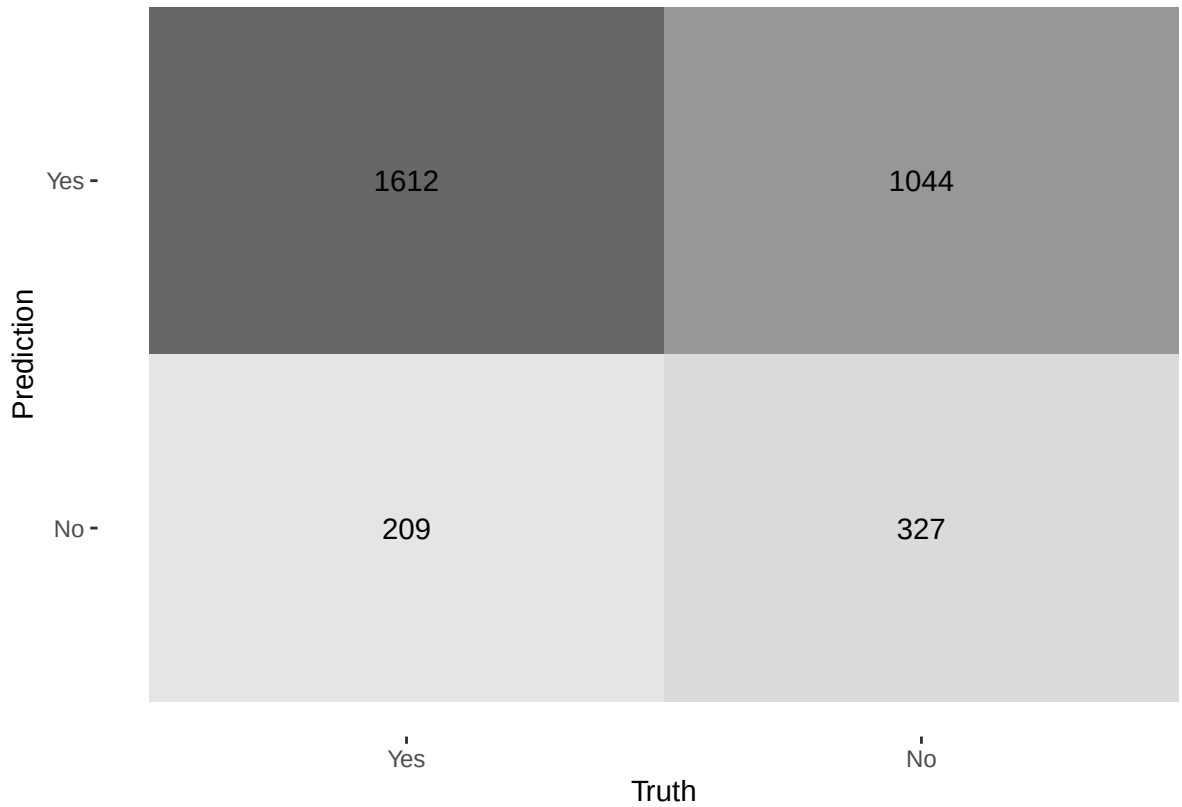
```
## # A tibble: 3 x 4
##   .metric     .estimator .estimate .config
##   <chr>       <chr>          <dbl> <chr>
## 1 accuracy    binary         0.624 pre0_mod0_post0
## 2 roc_auc     binary         0.648 pre0_mod0_post0
## 3 brier_class binary         0.234 pre0_mod0_post0
```

Confusion matrix for lda

```r
lda_preds <- collect_predictions(lda_fit)

# Confusion matrix
lda_preds %>%
  conf_mat(truth = high_achiever, estimate = .pred_class)
```

```
##           Truth
## Prediction  Yes   No
##        Yes 1612 1044
##        No   209  327
```

```r
# Heatmap
lda_preds %>%
  conf_mat(truth = high_achiever, estimate = .pred_class) %>%
  autoplot(type = "heatmap")
```

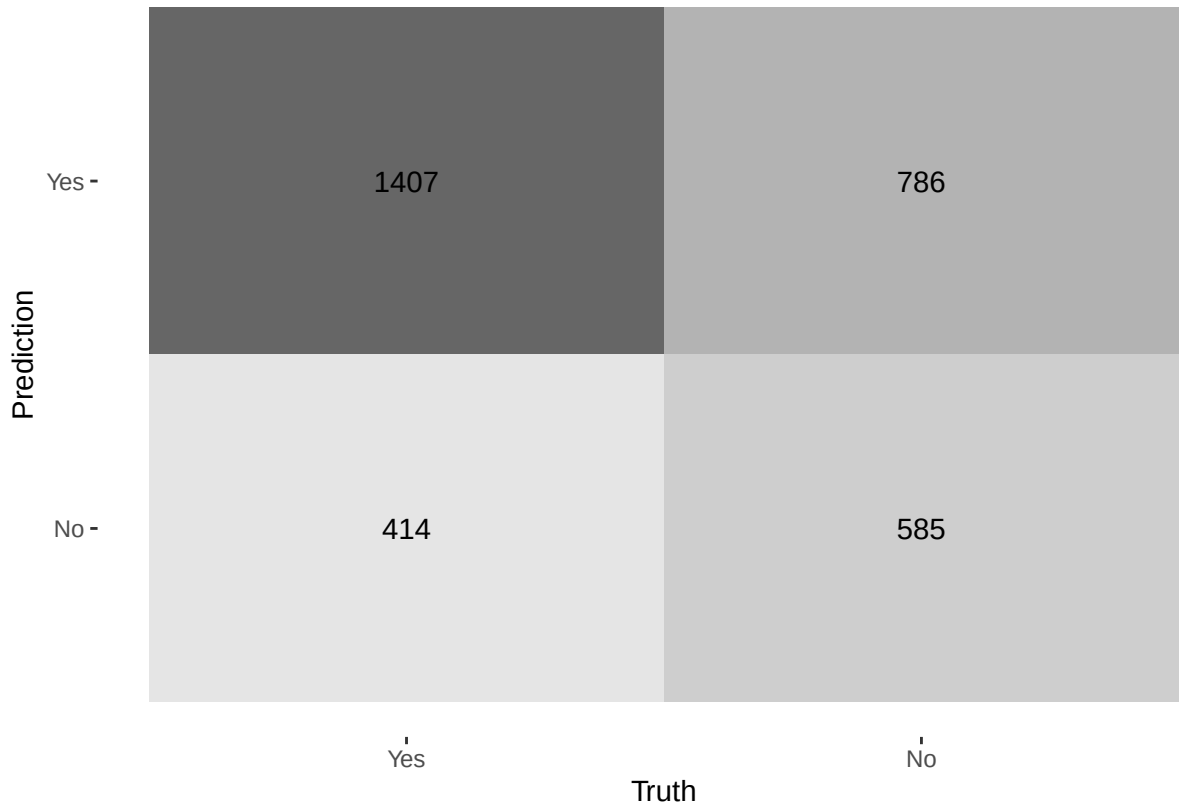


Confusion matrix for qda

```r
qda_preds <- collect_predictions(qda_fit)

qda_preds %>%
  conf_mat(truth = high_achiever, estimate = .pred_class)
```

```
##           Truth
## Prediction  Yes   No
##        Yes 1407  786
##        No   414  585
```

```r
qda_preds %>%
  conf_mat(truth = high_achiever, estimate = .pred_class) %>%
  autoplot(type = "heatmap")
```




```
metric_set(accuracy, precision, recall, f_meas, roc_auc)(
  lda_preds,
  truth = high_achiever,
  estimate = .pred_class,
  .pred_Yes
)
```

```
## # A tibble: 5 x 3
##   .metric   .estimator .estimate
##   <chr>     <chr>          <dbl>
## 1 accuracy  binary         0.607
## 2 precision binary         0.607
## 3 recall    binary         0.885
## 4 f_meas    binary         0.720
## 5 roc_auc   binary         0.617
```

```
metric_set(accuracy, precision, recall, f_meas, roc_auc)(
  qda_preds,
  truth = high_achiever,
  estimate = .pred_class,
  .pred_Yes
)
```

```
## # A tibble: 5 x 3
##   .metric   .estimator .estimate
##   <chr>     <chr>          <dbl>
## 1 accuracy  binary         0.624
```

Table 4: Poisson Dispersion Check for FSFREQ

mean_fsfreq	var_fsfreq	ratio
7.190098	92.28142	12.83451

```
## 2 precision binary         0.642
## 3 recall    binary         0.773
## 4 f_meas    binary         0.701
## 5 roc_auc   binary         0.648
```

Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA) were fit as alternative binary classifiers for high academic achievement. LDA achieved 60.7% accuracy, 88.5% recall, and an ROC AUC of 0.617, while QDA achieved 62.4% accuracy, 77.3% recall, and an ROC AUC of 0.648. LDA showed the highest recall, indicating that it was more aggressive in identifying positive cases, but this came at the cost of lower precision and weaker overall discrimination.

QDA performed slightly better than LDA in accuracy, precision, and AUC, suggesting that allowing quadratic decision boundaries improved classification somewhat. However, both discriminant-analysis methods underperformed compared with the logistic and lasso logistic models, indicating that logistic-based approaches were better suited to this dataset.

Poisson Regression Model

Since `FSFREQ` is literally a count of how many times adults in the household went to meetings or activities at the child's school, it is an excellent Poisson outcome.

```r
count_train %>%
  summarise(
    mean_fsfreq = mean(FSFREQ),
    var_fsfreq = var(FSFREQ),
    ratio = var(FSFREQ) / mean(FSFREQ)
  ) %>%
  kbl(caption = "Poisson Dispersion Check for FSFREQ") %>%
  kable_classic(full_width = FALSE)
```

If variance is much larger than the mean, we will note overdispersion. This will mean Poisson was demonstrated but may not fit perfectly.

```r
pois_recipe <- recipe(
  FSFREQ ~ school_involve_score + school_contact_score + FHHOME + FHWKHRS +
    FODINNERX + PARGRADEX + TTLHHINC + RACEETH + CSEX + DSBLTY + SCCHOICE + EDCPUB,
  data = count_train
) %>%
  step_dummy(all_nominal_predictors()) %>%
  step_zv(all_predictors())

pois_spec <- poisson_reg() %>%
  set_engine("glm") %>%
  set_mode("regression")

pois_wf <- workflow() %>%
  add_recipe(pois_recipe) %>%
  add_model(pois_spec)
```

```r
pois_fit <- fit(pois_wf, data = count_train)

pois_preds <- predict(pois_fit, new_data = count_test) %>%
  bind_cols(count_test)

rmse(pois_preds, truth = FSFREQ, estimate = .pred)
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>          <dbl>
## 1 rmse    standard        8.53
```

The Poisson regression model was used to analyze FSFREQ, the number of times adults in the household attended meetings or activities at the child's school. However, the dispersion check indicated substantial overdispersion: the mean of FSFREQ was 7.19, while the variance was 92.28, yielding a variance-to-mean ratio of 12.83. Because the Poisson model assumes equality of the mean and variance, this result indicates a serious violation of the Poisson assumption. The test-set RMSE of 8.53 was also large relative to the mean count, suggesting weak predictive accuracy. Therefore, while Poisson regression was demonstrated as required, it was not considered the preferred model for this count outcome.

Binary Model Comparison Table

```r
lasso_metrics <- metric_set(accuracy, precision, recall, f_meas, roc_auc)(
  lasso_preds, truth = high_achiever, estimate = .pred_class, .pred_Yes
) %>%
  select(.metric, .estimate) %>%
  pivot_wider(names_from = .metric, values_from = .estimate) %>%
  mutate(model = "Lasso Logistic")

log_metrics <- metric_set(accuracy, precision, recall, f_meas, roc_auc)(
  log_preds, truth = high_achiever, estimate = .pred_class, .pred_Yes
) %>%
  select(.metric, .estimate) %>%
  pivot_wider(names_from = .metric, values_from = .estimate) %>%
  mutate(model = "Standard Logistic")

lda_preds <- collect_predictions(lda_fit)
lda_metrics <- metric_set(accuracy, roc_auc)(
  lda_preds, truth = high_achiever, estimate = .pred_class, .pred_Yes
) %>%
  select(.metric, .estimate) %>%
  pivot_wider(names_from = .metric, values_from = .estimate) %>%
  mutate(model = "LDA")

qda_preds <- collect_predictions(qda_fit)
qda_metrics <- metric_set(accuracy, roc_auc)(
  qda_preds, truth = high_achiever, estimate = .pred_class, .pred_Yes
) %>%
  select(.metric, .estimate) %>%
  pivot_wider(names_from = .metric, values_from = .estimate) %>%
  mutate(model = "QDA")

bind_rows(lasso_metrics, log_metrics, lda_metrics, qda_metrics) %>%
```

Table 5: Binary Classification Model Comparison

model	accuracy	precision	recall	f_meas	roc_auc
Lasso Logistic	0.7149123	0.7265042	0.8023064	0.7625261	0.7734983
Standard Logistic	0.7077068	0.7231156	0.7902252	0.7551824	0.7688812
LDA	0.6074561	NA	NA	NA	0.6173456
QDA	0.6240602	NA	NA	NA	0.6480891

```
  select(model, everything()) %>%
  kbl(caption = "Binary Classification Model Comparison") %>%
  kable_classic(full_width = FALSE)
```

All-model comparison table

```
lasso_summary <- tibble(
  model = "Lasso Logistic",
  outcome = "high_achiever",
  accuracy = 0.7149123,
  roc_auc = 0.7734983,
  comment = "Best binary model"
)

log_summary <- tibble(
  model = "Standard Logistic",
  outcome = "high_achiever",
  accuracy = 0.7077068,
  roc_auc = 0.7688812,
  comment = "Close second"
)

lda_summary <- tibble(
  model = "LDA",
  outcome = "high_achiever",
  accuracy = 0.6074561,
  roc_auc = 0.6173456,
  comment = "High recall, weak AUC"
)

qda_summary <- tibble(
  model = "QDA",
  outcome = "high_achiever",
  accuracy = 0.6240602,
  roc_auc = 0.6480891,
  comment = "Better than LDA, below logistic"
)

multi_summary <- tibble(
  model = "Multinomial Regression",
  outcome = "grade_multi",
  accuracy = 0.6237469,
  roc_auc = 0.7119616,
```

Table 6: Model Comparison Summary

model	outcome	accuracy	roc_auc	comment
Lasso Logistic	high_achiever	0.715	0.773	Best binary model
Standard Logistic	high_achiever	0.708	0.769	Close second
Multinomial Regression	grade_multi	0.624	0.712	Best full-outcome model
LDA	high_achiever	0.607	0.617	High recall, weak AUC
QDA	high_achiever	0.624	0.648	Better than LDA, below logistic
Poisson Regression	FSFREQ	NA	NA	Rejected due to overdispersion; RMSE = 8.53

```
  comment = "Best full-outcome model"
)

pois_summary <- tibble(
  model = "Poisson Regression",
  outcome = "FSFREQ",
  accuracy = NA_real_,
  roc_auc = NA_real_,
  comment = "Rejected due to overdispersion; RMSE = 8.53"
)

bind_rows(
  lasso_summary, log_summary, multi_summary,
  lda_summary, qda_summary, pois_summary
) %>%
  kbl(
    caption = "Model Comparison Summary",
    digits = 3
  ) %>%
  kable_classic(full_width = FALSE)
```

- Because the grade-based outcomes are imbalanced, ROC AUC is emphasized more heavily than confusion-matrix accuracy when identifying the best classification models. Accuracy is still reported for completeness, but it can be misleading when most observations belong to the majority classes.
- The lasso model was used as a variable-selection tool to identify the most important family involvement, school contact, and demographic predictors of high academic achievement. Because lasso shrinks weak coefficients to zero, it provides a parsimonious starting point for final model interpretation.
- The standard logistic model provides interpretable odds ratios. Predictors with odds ratios greater than 1 are associated with higher odds of being a high achiever, holding other variables constant.
- The multinomial model is the most direct model for the original `SEGRADES` response because it preserves all four academic performance categories rather than collapsing them into a binary outcome.
- Poisson regression was used to model the count outcome `FSFREQ`, the number of times adults in the household attended meetings or activities at the child's school. This demonstrates the required count-modeling component of the project while remaining substantively tied to family involvement.

Preferred Models

Because the binary outcome `high_achiever` is imbalanced, model comparison emphasized ROC AUC in addition to confusion-matrix-based accuracy. Among the binary classifiers, the lasso logistic model is preferred because it achieved the strongest overall discrimination while also producing a more parsimonious set of

predictors. The standard logistic model performed similarly and remains useful for interpretation, but lasso had a slight edge in predictive performance. For the original four-category `SEGRADES` outcome, the multinomial regression model is preferred because it preserves the full academic-performance structure rather than collapsing the response into a binary indicator. LDA and QDA were useful benchmark methods, but both underperformed relative to the logistic-based models. Poisson regression was demonstrated for the count outcome `FSFREQ`, but it was not retained as a preferred model because strong overdispersion indicated poor fit.

Limitations

This analysis used a curated PFI analytic file and unweighted tidymodels workflows. The NHES:2019 Parent and Family Involvement survey is a complex national survey with weighting, nonresponse adjustment, and imputation procedures, so these results should be interpreted as associations within the analytic sample rather than fully survey-weighted national estimates. In addition, because the data are observational and cross-sectional, the models identify predictive associations rather than causal effects. For example, higher school involvement may reflect reactive engagement when a student is already struggling, rather than a direct causal improvement in grades.

Recommendations for GVSU's K-12 Connect

Across the exploratory analysis and the fitted models, home-based family engagement showed a clearer and more consistent relationship with academic performance than school-based involvement measures. In particular, homework-related engagement variables remained important in both the lasso-selected and regression-based models, whereas school event participation showed weaker and less consistent effects. Based on these findings, GVSU's K-12 Connect should prioritize programs that help families build effective homework routines, maintain regular academic support at home, and strengthen day-to-day engagement with children's schoolwork. School-based family involvement should still be encouraged, but the results suggest it may be more reactive than predictive when considered on its own.